

# torch.count\_nonzero#

[https://pytorch.org/docs/stable/generated/torch.count\\_nonzero.html](https://pytorch.org/docs/stable/generated/torch.count_nonzero.html)

## Description:

Counts the number of non-zero values in the tensor input along the given dim. If no dim is specified then all non-zeros in the tensor are counted.

input (Tensor) – the input tensor. dim (int or tuple of ints, optional) – Dim or tuple of dims along which to count non-zeros.

## Function Signature

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```
torch.count_nonzero(input, dim=None) → Tensor#
```

Parameters

## Code Examples

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```
>>> x = torch.zeros(3,3)
>>> x[torch.randn(3,3) > 0.5] = 1
>>> x
tensor([[0., 1., 1.],
        [0., 0., 0.],
        [0., 0., 1.]])
>>> torch.count_nonzero(x)
tensor(3)
>>> torch.count_nonzero(x, dim=0)
tensor([0, 1, 2])
```

# Detailed Documentation

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